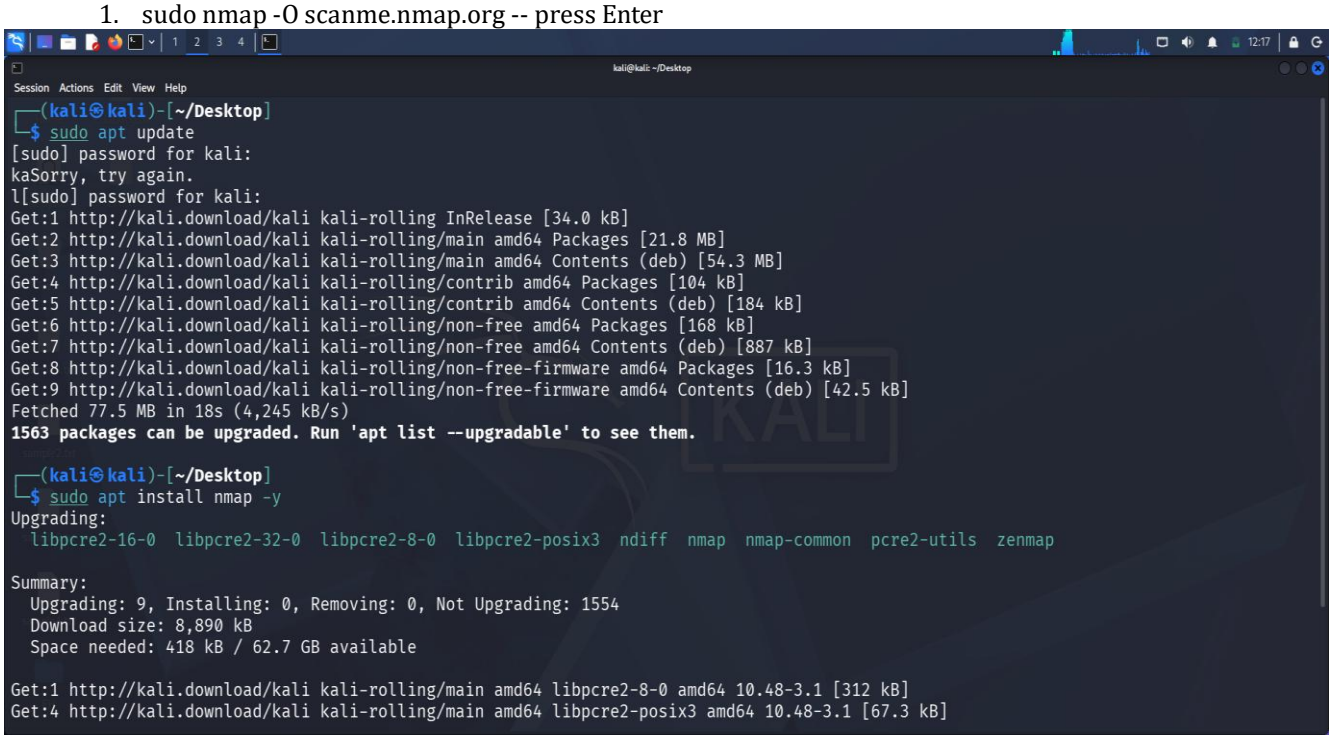
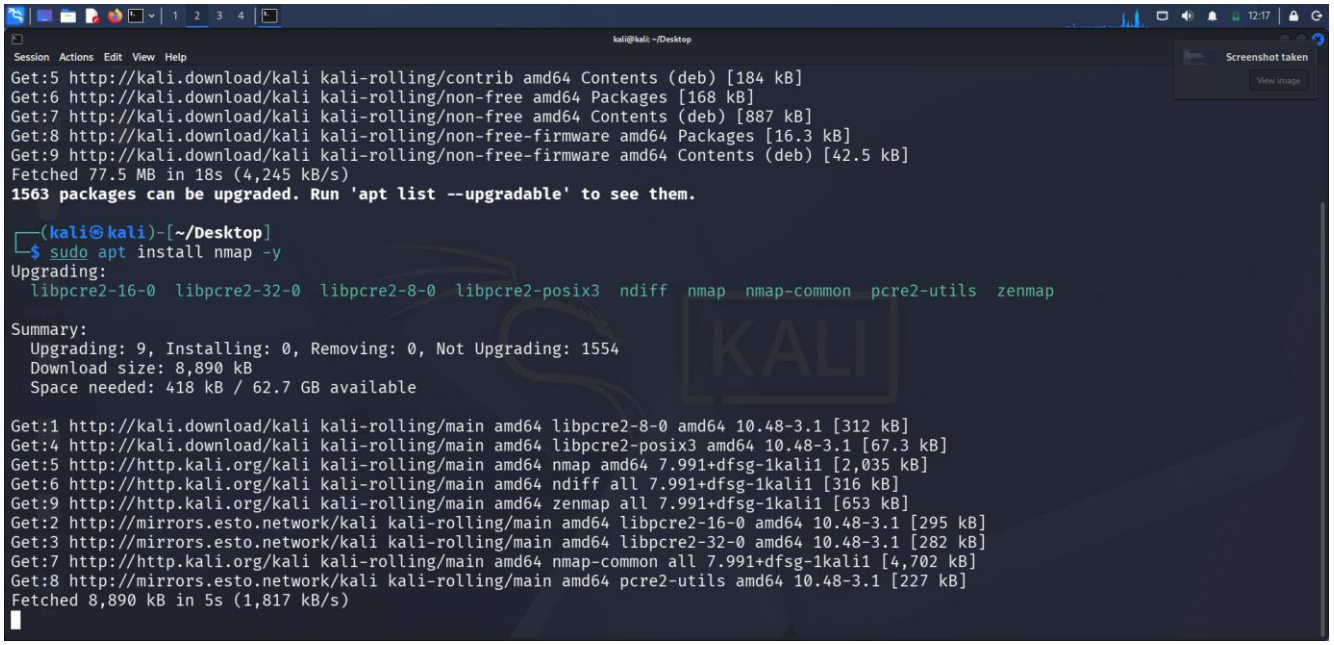
# ITA14 - ETHICAL HACKING
# EX - 8

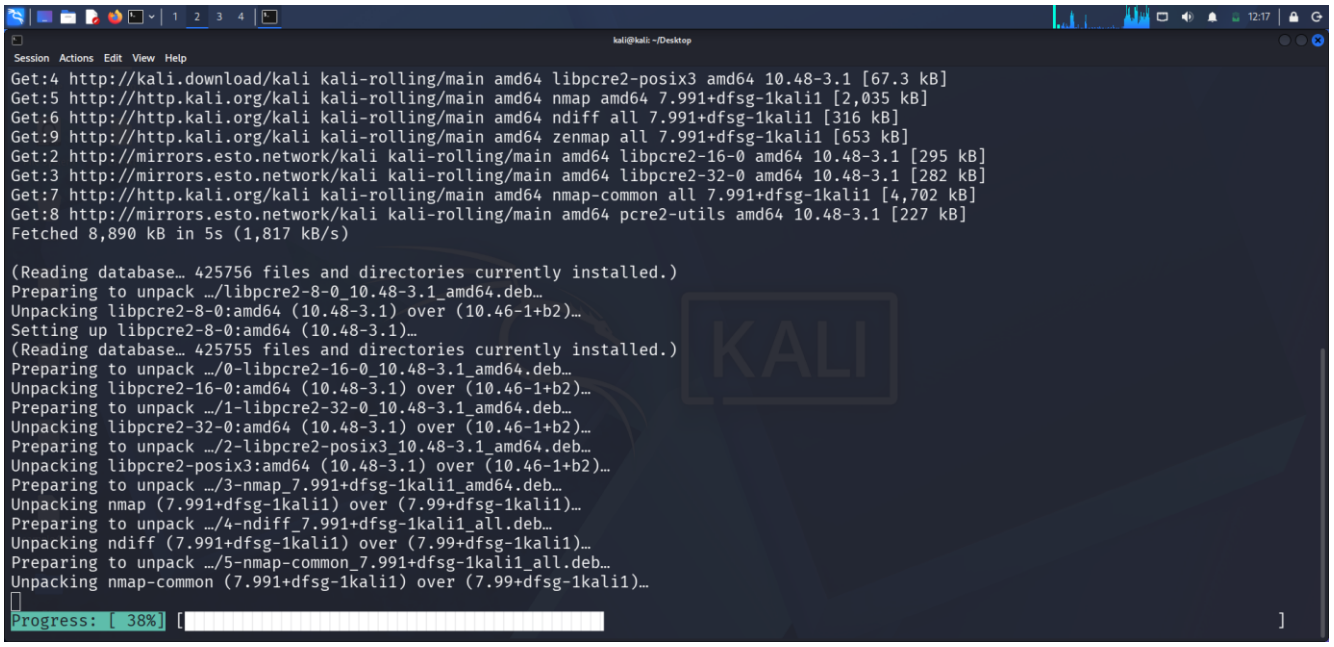
**Execute and display the results for the following Nmap service version and OS detection commands in Kali Linux operating systems:**
**i) detect the version of services running**
**ii) Aggressive scan**
**iii) Detect operating system of the target**

**Aim**
To execute and display the results for the following Nmap service version and OS detection commands in Kali Linux operating systems: i) detect the version of services running ii) aggressive scan iii) detect operating system of the target

**Procedure**
- Open (Cloud Shell Editor)
- i) detect the version of services running
    1. sudo apt update -- press Enter
    2. sudo apt install nmap -y -- press Enter
    3. nmap --version -- press Enter
    4. nmap -sV scanme.nmap.org -- press Enter
- ii) aggressive scan
    1. nmap -A scanme.nmap.org -- press Enter
- iii) detect operating system of the target
    1. sudo nmap -O scanme.nmap.org -- press Enter




Session Actions Edit View Help
Screenshot taken
Get:5 http://kali.download/kali kali-rolling/contrib amd64 Contents (deb) [184 kB]
Get:6 http://kali.download/kali kali-rolling/non-free amd64 Packages [168 kB]
Get:7 http://kali.download/kali kali-rolling/non-free amd64 Contents (deb) [887 kB]
Get:8 http://kali.download/kali kali-rolling/non-free-firmware amd64 Packages [16.3 kB]
Get:9 http://kali.download/kali kali-rolling/non-free-firmware amd64 Contents (deb) [42.5 kB]
Fetched 77.5 MB in 18s (4,245 kB/s)
1563 packages can be upgraded. Run 'apt list --upgradable' to see them.
┌──(kali㉿kali)-[~/Desktop]
└─$ sudo apt install nmap -y
Upgrading:
libpcre2-16-0 libpcre2-32-0 libpcre2-8-0 libpcre2-posix3 ndiff nmap nmap-common pcre2-utils zenmap
Summary:
Upgrading: 9, Installing: 0, Removing: 0, Not Upgrading: 1554
Download size: 8,890 kB
Space needed: 418 kB / 62.7 GB available
Get:1 http://kali.download/kali kali-rolling/main amd64 libpcre2-8-0 amd64 10.48-3.1 [312 kB]
Get:4 http://kali.download/kali kali-rolling/main amd64 libpcre2-posix3 amd64 10.48-3.1 [67.3 kB]
Get:5 http://http.kali.org/kali kali-rolling/main amd64 nmap amd64 7.991+dfsg-1kali1 [2,035 kB]
Get:6 http://http.kali.org/kali kali-rolling/main amd64 ndiff all 7.991+dfsg-1kali1 [316 kB]
Get:9 http://http.kali.org/kali kali-rolling/main amd64 zenmap all 7.991+dfsg-1kali1 [653 kB]
Get:2 http://mirrors.esto.network/kali kali-rolling/main amd64 libpcre2-16-0 amd64 10.48-3.1 [295 kB]
Get:3 http://mirrors.esto.network/kali kali-rolling/main amd64 libpcre2-32-0 amd64 10.48-3.1 [282 kB]
Get:7 http://http.kali.org/kali kali-rolling/main amd64 nmap-common all 7.991+dfsg-1kali1 [4,702 kB]
Get:8 http://mirrors.esto.network/kali kali-rolling/main amd64 pcre2-utils amd64 10.48-3.1 [227 kB]
Fetched 8,890 kB in 5s (1,817 kB/s)


Session Actions Edit View Help
Get:4 http://kali.download/kali kali-rolling/main amd64 libpcre2-posix3 amd64 10.48-3.1 [67.3 kB]
Get:5 http://http.kali.org/kali kali-rolling/main amd64 nmap amd64 7.991+dfsg-1kali1 [2,035 kB]
Get:6 http://http.kali.org/kali kali-rolling/main amd64 ndiff all 7.991+dfsg-1kali1 [316 kB]
Get:9 http://http.kali.org/kali kali-rolling/main amd64 zenmap all 7.991+dfsg-1kali1 [653 kB]
Get:2 http://mirrors.esto.network/kali kali-rolling/main amd64 libpcre2-16-0 amd64 10.48-3.1 [295 kB]
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Get:7 http://http.kali.org/kali kali-rolling/main amd64 nmap-common all 7.991+dfsg-1kali1 [4,702 kB]
Get:8 http://mirrors.esto.network/kali kali-rolling/main amd64 pcre2-utils amd64 10.48-3.1 [227 kB]
Fetched 8,890 kB in 5s (1,817 kB/s)
(Reading database… 425756 files and directories currently installed.)
Preparing to unpack …/libpcre2-8-0_10.48-3.1_amd64.deb…
Unpacking libpcre2-8-0:amd64 (10.48-3.1) over (10.46-1+b2)…
Setting up libpcre2-8-0:amd64 (10.48-3.1)…
(Reading database… 425755 files and directories currently installed.)
Preparing to unpack …/0-libpcre2-16-0_10.48-3.1_amd64.deb…
Unpacking libpcre2-16-0:amd64 (10.48-3.1) over (10.46-1+b2)…
Preparing to unpack …/1-libpcre2-32-0_10.48-3.1_amd64.deb…
Unpacking libpcre2-32-0:amd64 (10.48-3.1) over (10.46-1+b2)…
Preparing to unpack …/2-libpcre2-posix3_10.48-3.1_amd64.deb…
Unpacking libpcre2-posix3:amd64 (10.48-3.1) over (10.46-1+b2)…
Preparing to unpack …/3-nmap_7.991+dfsg-1kali1_amd64.deb…
Unpacking nmap (7.991+dfsg-1kali1) over (7.99+dfsg-1kali1)…
Preparing to unpack …/4-ndiff_7.991+dfsg-1kali1_all.deb…
Unpacking ndiff (7.991+dfsg-1kali1) over (7.99+dfsg-1kali1)…
Preparing to unpack …/5-nmap-common_7.991+dfsg-1kali1_all.deb…
Unpacking nmap-common (7.991+dfsg-1kali1) over (7.99+dfsg-1kali1)…
Progress: [ 38%] [


Session Actions Edit View Help
Setting up libpcre2-posix3:amd64 (10.48-3.1)…
Setting up nmap-common (7.991+dfsg-1kali1)…
Setting up ndiff (7.991+dfsg-1kali1)…
Setting up pcre2-utils (10.48-3.1)…
Setting up nmap (7.991+dfsg-1kali1)…
Setcap worked! Adding configuration to environment
Setting up zenmap (7.991+dfsg-1kali1)…
Processing triggers for mailcap (3.76)…
Processing triggers for kali-menu (2026.2.6)…
Processing triggers for desktop-file-utils (0.28-1)…
Processing triggers for hicolor-icon-theme (0.18-2)…
Processing triggers for libc-bin (2.42-16)…
Processing triggers for man-db (2.13.1-1)…
Processing triggers for libselinux1:amd64 (3.10-1)…
Processing triggers for wordlists (2026.2.0)…
┌──(kali㉿kali)-[~/Desktop]
└─$ nmap --version
Nmap version 7.991 ( https://nmap.org )
Platform: x86_64-pc-linux-gnu
Compiled with: liblua-5.4.9 openssl-3.6.2 libssh2-1.11.1 libz-1.3.2 libpcre2-10.48 libpcap-1.10.6 nmap-libdnet-1.18.0 ipv6
Compiled without:
Available nsock engines: epoll poll select
┌──(kali㉿kali)-[~/Desktop]
└─$ nmap -sV scanme.nmap.org
Starting Nmap 7.991 ( https://nmap.org ) at 2026-10-27 12:18 -0400
Warning: Hostname scanme.nmap.org resolves to 2 IPs. Using 45.33.32.156.